MAN1B1-Congenital Disorder of Glycosylation (Rafiq Syndrome): Comprehensive Disease Report

Disease: MAN1B1-Congenital Disorder of Glycosylation (MAN1B1-CDG; Rafiq syndrome, RAFQS) **MONDO ID:** MONDO:0018349 **Causal gene: MAN1B1** (mannosidase alpha class 1B member 1), chromosome 9q34.3 **Category:** Mendelian, autosomal recessive

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Summary

MAN1B1-Congenital Disorder of Glycosylation is a rare, autosomal-recessive, Golgi-type (type II) congenital disorder of glycosylation caused by **biallelic loss-of-function variants in the MAN1B1 gene** on chromosome 9q34.3. MAN1B1 encodes a glycoside-hydrolase family 47 (GH47) α-1,2-mannosidase that trims terminal α1,2-linked mannose residues from N-linked glycans and participates in endoplasmic reticulum (ER)/Golgi glycoprotein quality control. Loss of MAN1B1 function impairs N-glycan maturation, disrupts Golgi morphology, and produces a characteristic serum transferrin hybrid-type N-glycan "CDG-II" signature that is diagnostically useful. The disorder was first described clinically in 2010 and linked to MAN1B1 in 2011 by Rafiq and colleagues, giving rise to the eponym **Rafiq syndrome**.

Clinically, MAN1B1-CDG is dominated by a neurodevelopmental core: near-universal **intellectual disability/developmental delay** and **distinctive facial dysmorphism**, accompanied by **muscular hypotonia**, speech delay or absent speech, and behavioral problems (impulsivity, aggression). **Truncal obesity** — once highlighted in the alternative name "intellectual disability–truncal obesity syndrome" — is in fact one of the *least consistent* features across the ~45 patients reported worldwide. Additional, more variable features include epilepsy, cryptorchidism, feeding difficulties, hyperekplexia, and hypertransaminasemia (elevated liver transaminases). A recurrent missense allele, **c.1000C>T (p.Arg334Cys)**, appears repeatedly across unrelated populations.

No disease-modifying therapy exists. Management is entirely supportive and symptomatic, centered on developmental and behavioral support, monitoring of liver transaminases, and management of individual complications such as epilepsy. A pharmacologic "translational balancing" trial with disulfiram failed to alter glycosylation, reinforcing the absence of established targeted treatment. Despite the small absolute number of patients, MAN1B1-CDG is one of the more frequently encountered CDG-II subtypes in diagnostic cohorts — second only to PMM2 in some recent referral series — making accurate recognition of its biochemical and clinical signature clinically valuable.

Key Findings

Finding 1 — MAN1B1-CDG (Rafiq syndrome) is an autosomal-recessive CDG type II caused by biallelic MAN1B1 mutations

Exome sequencing identified **MAN1B1** as the causal gene in a previously unsolved CDG-II patient, and six additional cases were subsequently confirmed; all presented with slight facial dysmorphism, psychomotor retardation, and truncal obesity (PMID: 24348268). As the authors wrote: *"exome sequencing was used to identify MAN1B1 as the culprit gene in an unsolved CDG-II patient. Subsequently, 6 additional cases with MAN1B1-CDG were found. All individuals presented slight facial dysmorphism, psychomotor retardation and truncal obesity."*

The disorder was first described clinically in 2010 and genetically linked to MAN1B1 in 2011, and is now classified as **Rafiq syndrome (RAFQS)**, an autosomal-recessive CDG-II. A 2025 review of 44 cases confirmed the synonymy and inheritance pattern: *"Rafiq syndrome, MAN1B1-CDG, was described in 2010 and associated with genetic mutation in MAN1B1 gene in 2011. The disorder follows an autosomal recessive pattern of inheritance and typically presents with specific facial dysmorphism, intellectual disability, developmental delay, obesity, and hypotonia"* (PMID: 39840888). *(Evidence source: human clinical / exome sequencing.)*

Finding 2 — MAN1B1 encodes a Golgi-localized α-1,2-mannosidase; deficiency disrupts Golgi morphology and glycoprotein quality control

MAN1B1 was historically regarded as an ER-resident α-1,2-mannosidase initiating ER-associated degradation (ERAD). However, the index CDG study demonstrated that **endogenous MAN1B1 localizes to the Golgi complex**, and that patient cells show markedly abnormal

Golgi architecture: *"We were able to confirm that MAN1B1 is indeed localized to the Golgi complex instead of the ER. Furthermore, we observed an altered Golgi morphology in all patients' cells, with marked dilatation and fragmentation"* (PMID: 24348268).

Beyond its catalytic activity, MAN1B1 plays a **non-enzymatic "gatekeeper" role** in protein biosynthetic quality control. A conserved, vertebrate-specific non-enzymatic decapeptide in the luminal stem domain governs the fate of misfolded ERAD substrates: *"a highly conserved, vertebrate-specific non-enzymatic decapeptide sequence in the luminal stem domain plays a significant role in controlling the fate of overexpressed Null Hong Kong"* (PMID: 24627495). Mechanistically, MAN1B1 belongs to the **GH47 family**, which trims terminal α1,2-linked mannose residues during N-glycan maturation in the secretory pathway, producing the uniquely trimmed oligomannose isomers required for glycoprotein quality control and glycan maturation (PMID: 27856750). *(Evidence source: in vitro / cell biology / structural enzymology.)*

Finding 3 — MAN1B1-CDG produces a characteristic CDG-II serum transferrin glycoprofile with hybrid-type N-glycans, enabling diagnosis

High-resolution mass spectrometry of intact plasma transferrin reveals a highly characteristic glycosylation signature consisting of **hybrid-type N-glycans**, consistent with deficient mannosidase activity. Screening a cohort of 100 CDG-II patients with this method identified 11 additional MAN1B1-CDG cases, all confirmed by Sanger sequencing: *"A highly characteristic glycosylation signature was observed with hybrid type N-glycans, in agreement with deficient mannosidase activity. The speed and robustness of the method allowed subsequent screening in a cohort of 100 patients with congenital disorder of glycosylation type II, which revealed the characteristic glycosylation profile of MAN1B1-congenital disorder of glycosylation in 11 additional patients"* (PMID: 24566669).

In routine practice, serum transferrin isoelectric focusing shows a **type 2 CDG pattern**, and MALDI-TOF mass spectrometry of transferrin glycans aids diagnosis because the defective glycans accumulate only at trace levels (PMID: 34141584). A case series confirmed both the biochemical pattern and an associated laboratory abnormality: *"All patients had presented due to dysmorphic and neurological findings and hypertransaminasemia was remarkable. A type 2 pattern was found on serum transferrin isoelectrofocusing analysis"* (PMID: 34162022). *(Evidence source: human clinical / mass spectrometry diagnostics.)*

Finding 4 — Variant spectrum: biallelic loss-of-function with a recurrent p.Arg334Cys allele at locus 9q34.3

MAN1B1 maps to **chromosome 9q34.3** (the MRT15 intellectual-disability locus). The original gene-discovery study identified multiple loss-of-function variants (RefSeq NM_016219), including a nonsense variant **c.1418G>A (p.Trp473*)** segregating with intellectual disability in a Pakistani family, and demonstrated that disease-associated missense variants act through loss of function: *"we have identified mutations in the gene MAN1B1, encoding a mannosyl oligosaccharide, alpha 1,2-mannosidase. In one Pakistani family, MR43, a homozygous nonsense mutation (RefSeq number NM_016219.3: c.1418G>A [p.Trp473*]), segregated with intellectual disability"* (PMID: 21763484).

The recurrent missense allele **c.1000C>T (p.Arg334Cys)** has been reported repeatedly across unrelated populations and is functionally damaging: it either *"reduce[s] k(cat) by ~1300-fold or disrupt[s] stable protein expression in mammalian cells"* (PMID: 21763484). The expanding allelic spectrum now includes:

Variant (NM_016219)	Protein	Type	Population / note	Reference
c.1000C>T	p.Arg334Cys	Missense (recurrent)	Multiple populations; ~1300-fold ↓ kcat	PMID: 21763484, PMID: 39896699, PMID: 36142510
c.1418G>A	p.Trp473*	Nonsense	Pakistani family MR43	PMID: 21763484
c.1189G>A	p.Glu397Lys	Missense (founder)	Village founder allele	PMID: 21763484
c.1118C>G	p.Pro373Arg	Missense	Saudi	PMID: 41246981
c.1976T>G	p.Phe659Cys	Missense	Palestinian	PMID: 39840888
c.2011C>T + 23-bp deletion	—	Compound het (nonsense + del)	Chinese	PMID: 41393301
c.1065+1G>C	—	Splice-site	Russian (compound het with p.Arg334Cys)	PMID: 36142510

Both recurrent missense residues are conserved across the animal kingdom, and MAN1B1 has been cited as one of the few non-syndromic autosomal-recessive intellectual-disability (NS-ARID) genes with an elevated mutation frequency across populations. *(Evidence source: human genetics / in vitro functional assays.)*

Finding 5 — Clinical phenotype: neurodevelopmental core with variable truncal obesity; no disease-modifying therapy

Across the ~45 reported patients, the most frequent features are **intellectual disability / developmental delay** and **distinctive facial dysmorphism** (near-universal), while **truncal obesity is among the least consistent** features — a notable correction to the older syndrome name. The 44-case review stated plainly: *"the 2 most frequent features among them are intellectual disability and facial dysmorphism, whereas the least frequent one is truncal obesity"* (PMID: 39840888).

Additional recurrent features include muscular hypotonia, speech delay/absent speech, behavioral problems (impulsivity, aggression), and in subsets epilepsy, cryptorchidism, feeding difficulties, hyperekplexia, and hypertransaminasemia (PMID: 24348268; PMID: 39896699; PMID: 34162022). Novel phenotypic expansions reported include persistent hyperekplexia and feeding difficulty (PMID: 39896699) and co-occurring thyroid dyshormonogenesis in a case with a second-gene (DUOX2) variant (PMID: 41393301).

No targeted therapy exists. A trial of **disulfiram**, based on a translational-balancing hypothesis, left glycosylation unaltered — i.e., no biochemical benefit: *"Unaltered glycosylation during disulfiram treatment in mannosyl-oligosaccharide alpha-1,2-mannosidase-congenital disorders of glycosylation (MAN1B1-CDG)"* (PMID: 34258140). Management therefore remains supportive and symptomatic. *(Evidence source: human clinical / interventional trial.)*

Finding 6 — MAN1B1-CDG is one of the more frequently encountered CDG type II subtypes among diagnostic referrals

Although rare in absolute terms, MAN1B1-CDG is relatively common within CDG-II diagnostic populations. In the index diagnostic screen, 12 of 100 CDG-II patients (the index case plus 11 identified by transferrin glycoprofiling) carried MAN1B1 mutations (PMID: 24566669). In a 2022–2025 retrospective of biochemical genetic testing performed after equivocal molecular results, MAN1B1 was the **second most frequently implicated CDG gene after PMM2**: *"Molecular findings were reported for 52 different genes, of which PMM2 (n = 13), MAN1B1 (n =*

6), and ALG13 (n = 6) were the most frequent" (PMID: 41905312). Despite this relative frequency, the absolute number of reported patients remains small (~45 worldwide as of 2025). *(Evidence source: human clinical / diagnostic cohorts.)*

Section-by-Section Report

1. Disease Information

MAN1B1-CDG (Rafiq syndrome) is a rare autosomal-recessive inborn error of metabolism belonging to the **congenital disorders of glycosylation, type II (CDG-II)** — disorders characterized by defective processing of protein-bound N-glycans in the ER/Golgi. The disease arises from deficiency of the GH47 α-1,2-mannosidase encoded by MAN1B1, impairing N-glycan maturation and glycoprotein quality control.

Key identifiers: - **MONDO:** MONDO:0018349 - **Gene (HGNC):** MAN1B1 - **Locus:** 9q34.3 - **Synonyms:** MAN1B1-CDG; Rafiq syndrome (RAFQS); carbohydrate-deficient glycoprotein syndrome type II due to MAN1B1 deficiency; congenital disorder of glycosylation type 2/II due to MAN1B1 deficiency; intellectual disability–truncal obesity syndrome.

Information is derived from **aggregated disease-level resources and published case reports/series** (OMIM, Orphanet, primary literature), not from individual EHR records. The evidence base is a small, globally distributed set of ~45 reported patients.

2. Etiology

Primary cause — genetic. The disease is monogenic: biallelic (homozygous or compound heterozygous) **loss-of-function variants in MAN1B1** cause the disorder (PMID: 21763484). There are no established environmental or infectious causes.

Genetic risk factors. The sole causal factor is biallelic MAN1B1 pathogenic variation. **Consanguinity** is a strong contributor — many reported families are consanguineous (e.g., Pakistani, Saudi, Palestinian, and other populations), consistent with autosomal-recessive inheritance and homozygous founder/recurrent alleles.

Environmental risk / protective / gene–environment factors. No environmental risk factors, protective factors, or gene–environment interactions have been established for this Mendelian disorder. The only "protective" consideration is at the population-genetics level — avoidance of consanguineous unions reduces recessive-disease risk generally.

3. Phenotypes

Phenotype	Type	Frequency	HPO suggestion
Intellectual disability	Cognitive/ behavioral	Near-universal	HP:0001249
Global developmental delay	Developmental	Near-universal	HP:0001263
Distinctive facial dysmorphism	Physical	Near-universal	HP:0001999
Muscular hypotonia	Clinical sign	Frequent	HP:0001252
Speech delay / absent speech	Developmental	Frequent	HP:0000750
Behavioral problems (impulsivity, aggression)	Behavioral	Frequent	HP:0000708
Truncal obesity	Physical	Variable / least consistent	HP:0001956
Epilepsy / seizures	Neurological	Subset	HP:0001250
Hypertransaminasemia	Laboratory abnormality	Subset	HP:0002910
Cryptorchidism	Physical	Subset	HP:0000028
Feeding difficulties	Clinical sign	Subset	HP:0011968
Hyperekplexia (exaggerated startle)	Neurological	Rare/novel	HP:0002267

Characteristics. Onset is **congenital/neonatal to early childhood** (hypotonia, developmental delay recognized in infancy). Severity is **variable**, dominated by stable (non-degenerative) intellectual disability. The course is generally **static/non-progressive** neurologically, though behavioral issues and obesity may evolve. Frequency data derive from the 44-case review (PMID: 39840888), which ranked intellectual disability and facial dysmorphism as most frequent and truncal obesity as least frequent.

Quality of life. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disorder. Impact is driven by lifelong intellectual disability, communication limitations, and behavioral challenges, all of which substantially affect daily functioning and caregiver burden.

4. Genetic / Molecular Information

Causal gene: MAN1B1 (mannosidase alpha class 1B member 1), 9q34.3; RefSeq NM_016219. Encodes a GH47-family α-1,2-mannosidase.

Pathogenic variants. Reported variants are predominantly **loss-of-function missense, nonsense, splice-site, and small deletion** alleles (see table under Finding 4). The recurrent **c.1000C>T (p.Arg334Cys)** reduces catalytic rate ~1300-fold or abolishes stable protein expression (PMID: 21763484). Variant classification follows ACMG/AMP; truncating variants are pathogenic, and recurrent missense alleles are supported by functional data. Population allele frequencies (gnomAD) are low, consistent with rarity; MAN1B1 is nonetheless noted as having elevated mutation frequency among NS-ARID genes.

Functional consequence: loss of function (reduced/abolished mannosidase activity and/or loss of the non-enzymatic gatekeeper function). **Somatic vs germline:** germline. **Modifier genes / epigenetics / chromosomal abnormalities:** none established specific to MAN1B1-CDG. (One Chinese case carried a coexisting DUOX2 variant causing independent thyroid dyshormonogenesis — a second disease rather than a modifier; PMID: 41393301.)

Gene/GO annotations: molecular function GO:0004571 (mannosyl-oligosaccharide 1,2-α-mannosidase activity); process GO:0006491 (N-glycan processing); GO:0036508 (protein α-1,2-demannosylation).

5. Environmental Information

Not applicable. No environmental factors, lifestyle factors, or infectious agents are implicated in this monogenic recessive disorder. Consanguinity (a demographic/social factor) increases the probability of homozygous affected offspring but is not an environmental disease cause per se.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function variants in MAN1B1** (e.g., p.Arg334Cys, p.Trp473*) result in loss of GH47 α-1,2-mannosidase activity and/or loss of stable MAN1B1 protein.* (Demonstrated — functional assays, PMID: 21763484.)*
2. Loss of MAN1B1 **leads to** failure to trim terminal α1,2-linked mannose residues during N-glycan maturation in the Golgi/ER. *(Demonstrated — enzymology, PMID: 27856750.)*

3. Impaired mannose trimming **results in** accumulation of aberrant **hybrid-type N-glycans** on secreted glycoproteins (e.g., serum transferrin). *(Demonstrated — mass spectrometry, PMID: 24566669.)*
4. In parallel, loss of the **non-enzymatic gatekeeper decapeptide function leads to** defective ERAD substrate retention/retrieval and disturbed glycoprotein quality control. *(Demonstrated in overexpression models — PMID: 24627495.)*
5. These defects **lead to** altered Golgi morphology — marked dilatation and fragmentation — in patient cells. *(Demonstrated — PMID: 24348268.)*
6. Global hypoglycosylation/aberrant glycosylation of many client proteins **is inferred to** disrupt neuronal development and function. *(Inferred — mechanism linking glycosylation defect to neurodevelopment not directly demonstrated.)*
7. This **results in** the clinical phenotype: intellectual disability, developmental delay, facial dysmorphism, hypotonia, behavioral problems, and variable truncal obesity; a branch to the **liver** produces hypertransaminasemia. *(Observed clinically.)*

Detail by category: - **Molecular pathway:** N-linked glycan biosynthesis/maturation in the secretory pathway (KEGG "N-Glycan biosynthesis"; Reactome "Asparagine N-linked glycosylation"). GH47 mannosidases produce trimmed oligomannose isomers needed for quality control. - **Cellular processes:** ER-associated degradation (ERAD), glycoprotein quality control, Golgi-based glycan processing. GO:0030433 (ubiquitin-dependent ERAD pathway), GO:0006491. - **Protein dysfunction:** loss of catalytic activity (missense reducing kcat ~1300-fold) and/or loss of stable protein expression; also loss of a non-enzymatic chaperone-like gatekeeper function. - **Subcellular compartments:** Golgi apparatus (GO:0005794) — primary site of endogenous MAN1B1; ER (GO:0005789) involvement via ERAD. - **Biochemical abnormality:** mannosyl-oligosaccharide α-1,2-mannosidase deficiency; accumulation of hybrid-type N-glycans. - **Molecular profiling:** diagnostic transferrin glycoprofile (hybrid-type N-glycans). Albumin, newly recognized as N-glycosylated, also shows site-specific alterations in MAN1B1-CDG, a potential diagnostic biomarker (PMID: 41713138).

Suggested terms: GO:0004571, GO:0006491, GO:0030433; cell types affected are broad (neurons CL:0000540, hepatocytes CL:0000182) given multisystem glycoprotein involvement.

7. Anatomical Structures Affected

- **Primary organ/system:** central nervous system (brain; UBERON:0000955) — intellectual disability, developmental delay, hypotonia, behavioral problems.

- **Secondary involvement:** liver (UBERON:0002107) — hypertransaminasemia; adipose/metabolic (truncal obesity); craniofacial structures (dysmorphism); gonads (cryptorchidism).
- **Body systems:** nervous, hepatic/digestive, endocrine/metabolic, musculoskeletal (hypotonia).
- **Tissue/cell level:** broadly affected secretory-pathway–dependent cells; neurons and hepatocytes clinically prominent.
- **Subcellular: Golgi apparatus** (GO:0005794) is the key compartment (dilatation and fragmentation in patient cells); ER (GO:0005789) via ERAD.
- **Localization/lateralization:** systemic and bilateral/symmetric (a metabolic disorder, not a focal lesion).

8. Temporal Development

- **Onset:** congenital/neonatal to early childhood; typically recognized in infancy via hypotonia and developmental delay.
- **Onset pattern:** chronic/insidious.
- **Progression:** neurologically **static/stable** (non-degenerative) intellectual disability; this contrasts with some other CDGs (e.g., PMM2-CDG with progressive cerebellar atrophy). Behavioral features and obesity may fluctuate or evolve.
- **Duration:** chronic, lifelong.
- **Remission:** none (no spontaneous or treatment-induced remission).
- **Critical periods:** early developmental window relevant for supportive/rehabilitative intervention, though no disease-modifying therapy exists.

9. Inheritance and Population

- **Inheritance:** autosomal recessive (PMID: 39840888).
- **Penetrance:** complete (biallelic LoF produces disease); **expressivity:** variable (notably for truncal obesity and seizures).
- **Epidemiology:** no formal prevalence/incidence figures; ~45 patients reported worldwide as of 2025 (ultra-rare). Relatively common *within* CDG-II diagnostic cohorts — 12/100 in the index screen (PMID: 24566669) and the second most frequent CDG gene (n=6) after PMM2 (n=13) in a recent referral series (PMID: 41905312).

- **Founder effects/consanguinity:** recurrent/founder alleles (e.g., p.Glu397Lys village founder; recurrent p.Arg334Cys) reported; consanguinity frequent in affected families.
- **Populations:** reported across Pakistani, Saudi, Palestinian, Russian, Chinese, Korean, Portuguese, Turkish, and other populations — broad geographic distribution.
- **Sex ratio / age distribution:** no established sex bias (autosomal); affected individuals span pediatric through adult ages.
- **Carrier frequency / anticipation / mosaicism:** no specific data; anticipation not expected (not a repeat-expansion disorder).

10. Diagnostics

- **Screening biochemistry:** serum transferrin isoelectric focusing shows a **CDG type 2 pattern** (PMID: 34162022).
- **Confirmatory glycomics:** intact-transferrin high-resolution MS and MALDI-TOF MS of transferrin N-glycans reveal the characteristic **hybrid-type N-glycan** signature (PMID: 24566669; PMID: 34141584). Because defective glycans accumulate only at trace levels, a tiered decision tree (exome → gel electrophoresis → MS) improves diagnostic yield.
- **Emerging biomarker:** site-specific N-glycosylation changes in **albumin** (PMID: 41713138).
- **Laboratory clue:** hypertransaminasemia is a recurrent, supportive finding.
- **Genetic testing (definitive):** whole-exome sequencing has been the principal route to diagnosis (many cases are biochemically subtle); targeted/single-gene MAN1B1 sequencing and hereditary-metabolic gene panels also used. Chromosomal microarray/karyotype are not primary tools.
- **Differential diagnosis:** other CDG-II subtypes and non-specific syndromic intellectual disability; the hybrid-type glycan signature distinguishes MAN1B1-CDG. Caution: transferrin protein variants at glycosylation sites can mimic CDG patterns (PMID: 21431619).

11. Outcome / Prognosis

- **Survival/life expectancy:** no evidence of markedly reduced lifespan; the disorder is chronic and non-neurodegenerative, so prognosis is driven by the fixed intellectual disability and manageable complications rather than mortality. No formal survival data.
- **Morbidity/disability:** substantial lifelong disability from intellectual disability, communication limitation, and behavioral problems; obesity and epilepsy add morbidity in subsets.

- **Complications:** epilepsy, hypertransaminasemia/liver involvement, feeding difficulties, behavioral crises.
- **Prognostic factors:** severity of intellectual disability and presence of epilepsy/behavioral problems; no validated molecular prognostic biomarkers. Genotype–phenotype correlation is limited given the small cohort.

12. Treatment

- **No disease-modifying therapy exists.** A **disulfiram** trial based on a translational-balancing hypothesis did not alter glycosylation (PMID: 34258140).
- **Supportive/symptomatic management (standard of care):** early developmental intervention; physical, occupational, and speech therapy (NCIT rehabilitation interventions); behavioral support for impulsivity/aggression; **antiseizure medication** for epilepsy; nutritional/weight management for truncal obesity; and periodic monitoring of liver transaminases.
- **Advanced/experimental therapeutics:** none approved; no gene therapy, cell therapy, or RNA-based therapy in trials for MAN1B1-CDG.
- **Pharmacogenomics / targeted / immunotherapy:** not applicable.

(NCIT suggestions: Supportive Care; Physical Therapy; Occupational Therapy; Speech Therapy; Anticonvulsant Therapy.)

13. Prevention

- **Primary prevention:** not possible for a monogenic recessive disorder beyond reproductive risk reduction.
- **Genetic counseling:** central — carrier parents have a 25% recurrence risk per pregnancy; counseling on consanguinity-associated recessive risk is relevant.
- **Reproductive options:** carrier testing of at-risk relatives (cascade screening), prenatal diagnosis, and preimplantation genetic testing where a familial variant is known.
- **Newborn/population screening:** not currently part of standard newborn screening panels.
- **Secondary/tertiary prevention:** early diagnosis enables timely developmental support; monitoring and managing epilepsy, liver function, and obesity prevents complications.

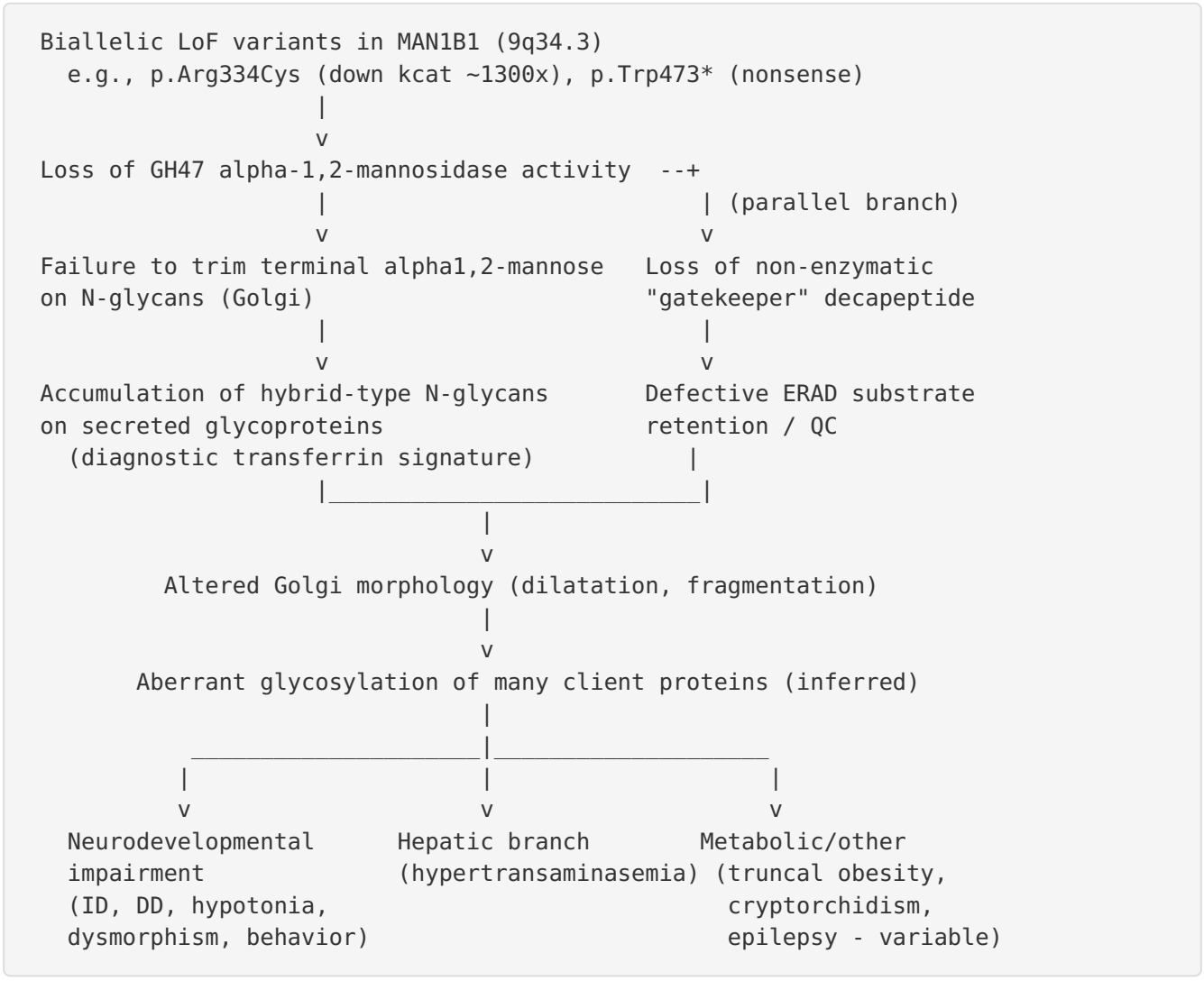
14. Other Species / Natural Disease

- **Taxonomy/orthologs:** MAN1B1 is conserved across vertebrates; the disease-relevant missense residues are conserved across the animal kingdom (PMID: 21763484), and the non-enzymatic gatekeeper decapeptide is vertebrate-specific (PMID: 24627495). Mouse ortholog: *Man1b1* (NCBI Gene ID 227619).
- **Natural disease in other species:** no naturally occurring MAN1B1-CDG analog documented in companion animals or wildlife (no OMIA entry identified).
- **Comparative biology:** high evolutionary conservation of the GH47 mannosidase mechanism supports cross-species functional relevance.
- **Transmission/zoonosis:** not applicable (genetic disorder).

15. Model Organisms

- **Direct MAN1B1-CDG animal models:** no well-characterized, published mammalian knockout model specifically recapitulating the human MAN1B1-CDG phenotype was identified in this investigation.
- **In vitro models:** patient-derived fibroblasts demonstrate altered Golgi morphology (dilatation/fragmentation) and the hybrid-type glycan defect, serving as the principal disease model (PMID: 24348268); mammalian cell overexpression systems established the loss-of-function mechanism of missense alleles and the gatekeeper role (PMID: 21763484; PMID: 24627495).
- **Related CDG models (context only):** other CDGs have animal models (e.g., Ngly1-deficient rats for NGLY1 deficiency, PMID: 41721346), but these model distinct genes, not MAN1B1.
- **Gap:** a conditional/knockout mammalian model would be a valuable resource (see Follow-up).

Mechanistic Model / Interpretation



MAN1B1-CDG illustrates two linked but distinct mechanisms converging on disease: a **catalytic defect** (impaired N-glycan mannose trimming) and a **non-catalytic quality-control defect** (loss of the gatekeeper function controlling misfolded-protein disposal). Upstream, the mutation and enzyme loss are well demonstrated; the biochemical consequence (hybrid-type glycans, abnormal Golgi) is directly observed in patient cells; the downstream link from generalized glycosylation disturbance to specific neurodevelopmental outcomes is plausible but remains **inferred** rather than mechanistically proven. The non-progressive clinical course and relatively mild biochemical signature (trace-level defective glycans) distinguish MAN1B1-CDG from more severe CDGs such as PMM2-CDG.

Evidence Base

PMID	Title (abbrev.)	Contribution
24348268	*MAN1B1 deficiency: an unexpected CDG-II*	Gene discovery via exome; core triad; Golgi localization + abnormal Golgi morphology
21763484	*Mutations in MAN1B1 cause AR intellectual disability*	9q34.3 locus; LoF variants; p.Arg334Cys down kcat ~1300x
24566669	*Diagnostic serum glycosylation profile (MAN1B1)*	Hybrid-type N-glycan transferrin signature; 12/100 CDG-II frequency
24627495	*Golgi-localized MAN1B1 non-enzymatic gatekeeper*	Non-enzymatic decapeptide gatekeeper role in ERAD/QC
27856750	*GH47 alpha-mannosidases and glycan maturation*	Enzymatic mechanism of GH47 mannose trimming
39840888	*Rafiq syndrome, Palestinian child, 44-case review*	AR inheritance; frequency ranking (ID/dysmorphism > obesity)
34162022	*MAN1B1-CDG: novel patients and variant*	CDG type 2 IEF pattern; hypertransaminasemia
34141584	*MAN1B1-CDG: three new individuals, biochemical profiles*	MALDI-TOF diagnostic decision tree; trace-level glycans
34258140	*Disulfiram translational balancing*	Failed pharmacologic trial — no disease-modifying therapy
39896699	*Rafiq syndrome, new phenotypic features*	p.Arg334Cys; feeding difficulty + hyperekplexia (novel)
41905312	*Biochemical genetic testing after equivocal sequencing*	MAN1B1 2nd most frequent CDG gene (n=6) after PMM2
41246981	*Saudi patient, p.Pro373Arg*	Novel variant; expanded phenotype
41393301	*Chinese patient + thyroid dyshormonogenesis*	Compound het variants; coexisting DUOX2 disease
36142510	*Russian patient, compound het*	c.1065+1G>C splice variant with p.Arg334Cys
41713138	*Albumin as CDG glycoprotein biomarker*	Albumin N-glycosylation altered in MAN1B1-CDG

Evidence quality is predominantly **human clinical case reports/series** (small n) plus **in vitro cell biology and enzymology**. The gene-disease relationship, loss-of-function mechanism, Golgi pathology, and diagnostic glycoprofile are well supported; genotype–phenotype correlations and the precise neurodevelopmental mechanism are not.

Limitations and Knowledge Gaps

1. **Small cohort (~45 patients):** prevents robust prevalence estimates, genotype–phenotype correlation, and natural-history/prognosis data.
2. **Inferred downstream mechanism:** the causal link from generalized glycosylation disturbance to specific neurodevelopmental and metabolic outcomes is plausible but not directly demonstrated.
3. **No animal model:** no published mammalian knockout model recapitulating MAN1B1-CDG limits mechanistic and preclinical therapeutic study.
4. **No quantitative QoL/outcome data:** no EQ-5D/SF-36/PROMIS or standardized disability metrics.
5. **No therapy:** only one pharmacologic trial (disulfiram), which was negative; no targeted therapeutic strategy.
6. **Diagnostic subtlety:** trace-level defective glycans make biochemical diagnosis difficult; exome sequencing is often required.
7. **Variant interpretation:** several alleles are recent/novel; functional validation is incomplete for some missense variants.

Proposed Follow-up Experiments / Actions

1. **Establish a conditional/knockout mammalian (mouse) model** of *Man1b1* deficiency to test phenotype recapitulation (neurodevelopment, obesity, liver), and a cellular iPSC-derived neuronal model to probe the neurodevelopmental mechanism.
2. **Build an international MAN1B1-CDG registry/natural-history study** to quantify phenotype frequencies, QoL, and longitudinal course across the ~45+ patients.

3. **Systematic genotype–phenotype analysis** correlating variant class (truncating vs missense; residual activity) with severity, especially for the recurrent p.Arg334Cys.
4. **Validate albumin-derived glycopeptides** and refine transferrin MS workflows as sensitive, specific diagnostic biomarkers to shorten the diagnostic odyssey (PMID: 41713138; PMID: 24566669).
5. **Functional characterization** of newly reported variants (p.Pro373Arg, p.Phe659Cys, splice-site alleles) in mammalian expression systems to confirm pathogenicity and ACMG classification.
6. **Preclinical therapeutic exploration:** given the dual catalytic + gatekeeper mechanism, test proteostasis-modulating and glycan-targeted strategies in cellular models; reassess any residual-function rescue approaches now that disulfiram has failed.
7. **Incorporate MAN1B1 into CDG gene panels and reflex glycomics** given its relative frequency among CDG-II referrals.
